

SANCHIT KABRA

✉ sanchit23@vt.edu  [Sanchit Kabra](#)  [Sanchit-404](#)

Research Focus

Self-Evolving Agents; Reasoning in Large Language Models; Red Teaming of LLM Agents; AI for Scientific Discovery

Education

Virginia Tech

PhD, MS (exp. 2026) in Computer Science

2024 – Present

Virginia, USA

Birla Institute of Technology and Science, Pilani

Bachelor of Engineering in Computer Science(Honors)

2020 – 2024

Pilani, India

Publications *Google Scholar*

- **Accelerating Materials Design via LLM-Guided Evolutionary Search.** Nikhil Abhyankar*, **Sanchit Kabra***, Saaketh Desai, Chandan K. Reddy. *ICLR 2026*
- **SURFACEBENCH: Can Self-Evolving LLMs Find the Equations of 3D Scientific Surfaces?.** Sanchit Kabra, Shobhnik Kriplani, Parshin Shojaei, Chandan K. Reddy. *TMLR*, 2026.
- **Decompose, Adapt and Evolve: Towards Efficient Scientific Equation Discovery with Large Language Models.** Pouya Behzadifar*, Parshin Shojaei*, **Sanchit Kabra***, Kazem Meidani, Chandan K. Reddy. *NeurIPS AI for Science Workshop*, 2025.
- **Reasoning Towards Fairness: Mitigating Bias in Language Models through Reasoning-Guided Fine-Tuning.** **Sanchit Kabra**, Akshita Jha, Chandan K. Reddy. *Preprint*, 2025.
- **Biased or Flawed? Mitigating Stereotypes in Generative Language Models by Addressing Task-Specific Flaws.** Akshita Jha, **Sanchit Kabra**, Chandan K. Reddy. *Preprint*, 2024.
- **RUST-BENCH: Benchmarking LLM Reasoning on Unstructured Text within Structured Tables.** Nikhil Abhyankar, Purvi Chaurasia, **Sanchit Kabra**, Ananya Srivastava, Vivek Gupta, Chandan K. Reddy. *Preprint*, 2025.
- **Aligned at the Start: Concept Formation in LLM Embeddings.** Mehrdad Khatir, **Sanchit Kabra**, Chandan K. Reddy. *Preprint*, 2024.

Research Experience

Virginia Tech – *Research Assistant*

Advisor: Chandan Reddy

May 2022 – Present

Arlington, VA

Red Teaming & Security Evaluation

- **LLMs as Evolutionary Attackers:** Developed a novel LLM-driven evolutionary framework for discovering single-turn jailbreak prompts under strict black-box query budgets.
 - Discovered **3× more unique successful jailbreaks** than template-based attacks under the same budget.
 - Achieved **100% ASR** and **82–86%** judge-validated ASR on **LLaMA-2-7B-Chat** and **GPT-4o-mini**, surfacing structurally distinct attack strategies.
- **Multi-Vulnerability Code Security Benchmarking:** Developed a novel file-level benchmark for multi-vulnerability detection across **C, C++, Python, and JavaScript**.
 - Generated long-context programs with multiple concurrent CWEs per file to evaluate LLM-based security auditing and identified severe performance degradation, with recall dropping by **30–50%** and Exact-File accuracy decreasing by up to **40%** in dense vulnerability settings.

AI for Science

- **Materials Discovery (LLEMA):** Developed an LLM-guided evolutionary search system for multi-objective materials discovery evaluated across **10 industrial-scale tasks**.
 - Achieved up to **2× higher hit-rates** than generative and LLM-only baselines, consistently discovering chemically valid and thermodynamically stable candidates.
 - Produced stronger Pareto fronts with higher hypervolume and earlier convergence across all tasks.
- **Scientific Equation Discovery (DecAEvolve):** Developed **DecAEvolve**, a novel adaptive Reinforcement Learning based evolutionary framework for LLM-based symbolic regression that adapts the hypothesis generator at test time.

- Reduced discovery error by **1.5–3× (normalized MSE)** compared to the state-of-the-art LLM-SR baseline under identical **3K-query budgets**.
- Achieved **1.5–2× faster convergence**, reaching comparable or lower error in **40–60% fewer iterations** and delivering the best out-of-distribution NMSE across **0.5B and 1.5B** models.

Responsible AI & Alignment

- **Bias Analysis and Mitigation in Generative Language Models:** Designed a novel framework to disentangle identity-conditioned bias from task-specific comprehension and reasoning failures in ambiguous language understanding.
- Conducted large-scale evaluations on **fairness** and **reading comprehension** datasets, showing that **> 80%** of stereotypical errors arise from reasoning failures rather than inherent stereotypical bias.
- Reduced stereotypical outputs by **60%+** through instruction tuning without using fairness specific data.
- **Reasoning-Guided Bias Mitigation:** Developed a novel reasoning-guided fine-tuning to distill structured reasoning traces into non-reasoning models, improving fairness accuracy from **40–50%** to **85–90%** across multiple fairness dimensions.

Aalto University – *Research Assistant*

June 2023 – Jan 2024

Advisor: Rohit Babbar

Finland

- Reformulated large-scale recommendation as an extreme multi-label **ranking and retrieval** problem, modeling each candidate item as a label, achieving **+12% F1** and **+8 pass@5** over collaborative filtering and content-based baselines
- Developed and trained a **multimodal retrieval model** using the LightXML pipeline on **5M+ samples** curated from the AmazonTitles-670K dataset, enabling scalable inference over a large item space

Projects

Amazon Trusted AI Challenge – Code-Based LLMs

October 2024 – June 2025

Advisor: Dr. Ruoxi Jia

Amazon / Virginia Tech

- Developing an **8B-parameter code-focused LLM** from scratch, including pretraining and finetuning, as part of one of the 5 blue teams at Amazon Trusted AI Challenge, ensuring robustness in code generation tasks
- Engineered a **Chain-of-Thought (CoT) knowledge distillation** framework to extract reasoning capabilities from larger models such as GPT-4-O1 and DeepSeek R1 and transfer them to an 8B parameter base model, improving its performance on complex programming tasks
- Designed adversarial and supervised finetuning pipelines to reduce the model's susceptibility to generating **malicious, insecure, or vulnerable code** while maintaining functional correctness and diversity in outputs
- Conducted large-scale evaluations on security benchmarks, leveraging static vulnerable code analysis tools along with dynamic LLM driven systems for evaluation of context specific security events.

Awards

- **Amazon Trusted AI Challenge, 2024:** Advisor: Rouxi Jia. A global university competition focused on responsible AI and large language model coding security. Part of **HokieTokie**, one of the 10 teams selected globally. Received **USD 900,000** AWS compute credits over the course of competition.
- **Higher Secondary School Exams State Topper, 2020:** Ranked second in the Maharashtra State Board Examinations, among 1.5 million students. Scored a perfect 100/100 in Math, Chemistry, Physics, and Biology.
- **INSPIRE Scholarship, 2020:** Offered by the Department of Science & Technology (DST), Government of India for finishing in the top 1% in Class XII Board Examinations.

Skills

- **Programming Languages:** Python, Java, R, Matlab, C/C++, SQL, bash
- **Frameworks and Tools:** PyTorch, TensorFlow, Numpy, Pandas, RAG, Huggingface, SciKit, Matplotlib, pyspark
- **Technologies:** AWS, Linux

Relevant Coursework

- | | | | |
|-------------------------------------|-------------------------|-------------------------------|-------------------------------------|
| • Data Structures and Algorithms | • Operating Systems | • Deep Learning | • Decision Making Under Uncertainty |
| • Design and Analysis of Algorithms | • Database Systems | • Natural Language Processing | • Computational Learning Theory |
| | • Compiler Construction | • Multimodal Vision | |
| | • Computer Architecture | | |

